

Megan K. Ayers

New Haven, CT | m.ayers@yale.edu

RESEARCH INTERESTS

Causal inference, environmental impact evaluation, experimental design, remote sensing, machine learning, text as data, computational social science, social and environmental applications of statistics

EDUCATION

Ph.D. in Statistics & Data Science, Yale University, Fall 2025 (expected)

M.A. in Statistics & Data Science, Yale University, 2024

B.A. in Mathematics, B.A. in Physics, Lewis & Clark College, 2019

PUBLICATIONS

PEER REVIEWED JOURNAL PUBLICATIONS

Ayers, M., Marlon, J. R., Ballew, M. T., Maibach, E. W., Rosenthal, S. A., Roser-Renouf, C., & Leiserowitz, A. (2024). Changes in Global Warming's Six Americas: An analysis of repeat respondents. *Climatic Change*, 177(6), 96. <https://doi.org/10.1007/s10584-024-03754-x>

Liu, T., Gezari, S., **Ayers, M.**, Burgett, W., Chambers, K., Hodapp, K., Huber, M. E., Kudritzki, R.-P., Metcalfe, N., Tonry, J., Wainscoat, R., & Waters, C. (2019). Supermassive Black Hole Binary Candidates from the Pan-STARRS1 Medium Deep Survey. *The Astrophysical Journal*, 884(1), 36. <https://doi.org/10.3847/1538-4357/ab40cb>

Foord, A., Gültekin, K., Reynolds, M., **Ayers, M.**, Liu, T., Gezari, S., & Runnoe, J. (2017). A Multi-wavelength Analysis of Binary-AGN Candidate PSO J334.2028+01.4075. *The Astrophysical Journal*, 851(2), 106. <https://doi.org/10.3847/1538-4357/aa9a39>

CONFERENCE & WORKSHOP PUBLICATIONS

Ayers, M.^{*}, Sanford, L.^{*}, Roberts, M. E., & Yang, E. (2024). Discovering influential text using convolutional neural networks. *Findings of the Association for Computational Linguistics*, pages 12002–12027, Bangkok, Thailand and virtual meeting.
^{*}Equal contribution

Gordon, M., **Ayers, M.**, Stone, E., & Sanford, L. (2023). Remote Control: Debiasing Remote Sensing Predictions for Causal Inference. *Proc. Int. Conf. Learn. Represent. Workshops*, Kigali, Rwanda and virtual meeting.

PREPRINTS

Ayers, M.^{*}, Sanford, L.^{*}, Gardner, W., & Kuebbing, S. (2025). Causal Carbon: Baselines and Additivity with Potential Outcomes. https://doi.org/10.31219/osf.io/5pcuh_v2
^{*}Equal contribution

Sanford, L.^{*}, **Ayers, M.**^{*}, Gordon, M.^{*}, & Stone, E. (2025). Adversarial Debiasing for Unbiased Parameter Recovery. <https://www.arxiv.org/abs/2502.12323>
^{*}Equal contribution

TEACHING & MENTORING EXPERIENCE

Teaching Fellow, Yale University, 2021-2024

- Statistical Case Studies (Fall 2022, Fall 2024)
- Data Analysis (Spring 2022, Spring 2023)
- Statistical Computing (Fall 2022)

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- Linear Models (Fall 2021)

Grader without Contact, Yale University, 2023-2025

- Statistical Consulting (Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025)

College Teaching Preparation Program Workshops, Yale University

- Advanced Teaching two-part workshop series on “How We Learn”, Spring 2025
- Advanced Teaching Workshop on Course Design, Fall 2024
- Intermediate Teaching Workshop on Teaching First-Generation and Low-Income Students, Fall 2024
- Intermediate Teaching Workshop on Grading and Rubrics, Fall 2024
- Intermediate Teaching Workshop on Mental Health in the Classroom, Fall 2024
- Intermediate Teaching Workshop on Teaching Quantitative Reasoning, Spring 2024
- Fundamentals of Equitable Teaching four-part workshop series, Spring 2024

Peer Tutor in Math & Physics, Lewis & Clark College, 2016-2019

Math Department Grader, Lewis & Clark College, 2016-2019

- Partial Differential Equations with Applications (Spring 2019)
- Linear Algebra (Fall 2018)
- Differential Equations (Fall 2017)
- Calculus I (Fall 2016, Spring 2017)

FELLOWSHIPS & AWARDS

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| 2023-2024 | Planetary Solutions Project seed grant from the Climate Impact Innovation Fund and the Gordon Data and Environmental Sciences Research Grant (research assistant funding recipient), Yale University |
| 2023 | Graduate Student Assembly Conference Travel Fellowship, Yale University |
| 2020 | Sterling Prize Fellowship, Yale University |
| 2018 | Shannon T. O’Leary Award, Lewis & Clark College |

PRESENTATIONS

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| 2024 | “Discovering influential text using convolutional neural networks,” Association for Computational Linguistics , August 12-14 and August 22, Bangkok, Thailand and virtual meeting. (Virtual poster) |
| | “A New Perspective for Forest Carbon Offsetting through the Potential Outcomes Framework,” American Causal Inference Conference Annual Meeting , May 14-17, Seattle, WA. (Poster) |
| | “Site Matching Baselines with Real and Synthetic Controls,” Workshop on Increasing Consensus on Best Practices for Additionality & Baselines in Forest Carbon Crediting Accounting , April 11-12, Yale University. (Oral) |
| 2023 | “Assessing Leakage from Carbon Offset Projects,” Google Geo for Good Summit , October 10-12, Mountain View, CA. (Oral) |
| | “Remote Control: Debiasing Remote Sensing Predictions for Causal Inference,” The Workshop in Environmental Economics and Data Science , September 15-16, Portland, OR. (Oral, presented by a colleague for health reasons) |
| | “Remote Control: Debiasing Remote Sensing Predictions for Causal Inference,” ICLR Workshop on Machine Learning for Remote Sensing , May 5, Kigali, Rwanda and |

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Virtual meeting. (Virtual oral)

“Remote Control: Debiasing Remote Sensing Predictions for Causal Inference,” **ICLR Workshop on Tackling Climate Change with Machine Learning**, May 4, Kigali, Rwanda and virtual meeting. (Virtual poster)

RESEARCH & PROFESSIONAL EXPERIENCE

Yale Center for Environmental Law & Policy
Summer Research Assistant

New Haven, CT
June-August 2022

Yale Law School Legal Services Organization
Statistical/Data Analyst

New Haven, CT
November 2020-March 2021

Cascade Data Labs
Data Analyst

Portland, OR
June 2019-May 2020

Yale University Sackler NSF-REU
Research Intern (Emonet Lab)

New Haven, CT
May-August 2018

Maria Mitchell Observatory NSF-REU
Research Intern

Nantucket, MA
May-August 2017

SERVICE

Ad-hoc reviewer, *Climatic Change*, 2024

Workshop organizer, Workshop on Increasing Consensus on Best Practices for Additionality & Baselines in Forest Carbon Crediting Accounting, 2024

Undergraduate Learning Assistant hiring coordinator, Yale University Department of Statistics & Data Science, 2021-2022